Logit Adjustment Meets Transfer Learning: Balanced Skin-Lesion Classification on DermaMNIST

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Abstract—We classify 28×28 dermatoscopy images of DermaMNIST into seven lesion types, two thirds of which are harmless moles, and optimise balanced accuracy. Starting from the starter CNN (validation balanced accuracy 0.461 ± 0.011 over three seeds), we tested ten single changes, each with three seeds and a 95 % paired bootstrap interval against the baseline. Correcting the class imbalance helped most: a logit-adjusted loss raised validation balanced accuracy to 0.532 ± 0.031. Combined with an ImageNet-pretrained ResNet-18 on upsampled inputs and flip/rotation augmentation, it reached 0.590 ± 0.017, and an ensemble of the three seed models reached 0.609. On the official test set, evaluated once, this model scores 0.632 balanced accuracy against 0.448 for the baseline ensemble (Δ +0.184, 95 % CI [+0.136, +0.232]). The gain comes from finding rare classes, at the price of more moles being called something else.

I. Introduction

A useful skin-lesion classifier must recognise every lesion type, not only the most common one. In DermaMNIST 67 % of the training images are melanocytic nevi, so a model that always answers "nevus" reaches 67 % accuracy but finds no melanoma; its balanced accuracy (mean sensitivity over the seven classes) is 1/7. Balanced accuracy weights every class equally, including rare benign ones such as dermatofibroma, so we select and report every model by it. Our contributions: (i) a controlled comparison of ten single changes to the starter baseline, each with three seeds and a bootstrap interval; (ii) a stack of the helpful changes, chosen on validation data only; (iii) evidence that the imbalance correction and the pretrained network help each other.

II. Data

DermaMNIST [1] contains the HAM10000 dermatoscopy images [2] resized to 28×28 RGB, with the official split of 7 007 training, 1 003 validation and 2 005 test images (Table I). The imbalance is 59:1 between nevi (4 693) and dermatofibroma (80); the validation set holds only 12 dermatofibroma and 14 vascular images, so their sensitivities change by 7–8 points per image. Images are scaled to [0,1] and normalised with the starter's per-channel mean (0.7631, 0.5381, 0.5614) and std (0.1366, 0.1543, 0.1692); augmentation, where used, is applied to training images only. The 28×28 resolution removes fine texture and borders, and HAM10000 can contain several images of the same lesion; we use the official splits as given and treat this as a limitation.

TABLE I
Images per class and split

label	class	train	val	test
akiec	actinic keratoses†	228	33	66
bcc	basal cell carcinoma	359	52	103
bkl	benign keratosis-like lesions	769	110	220
df	dermatofibroma	80	12	23
mel	melanoma	779	111	223
nv	melanocytic nevi	4 693	671	1 341
vasc	vascular lesions	99	14	29

† and intraepithelial carcinoma

III. Methods

A. Baseline (A)

The starter CNN has two blocks of 3×3 convolution (16 and 32 channels), ReLU and 2×2 max-pooling, followed by a linear layer on the 1 568 flattened features (16 071 parameters). It is trained with cross-entropy and Adam [3] (learning rate 10^{-3}, batch size 64) for 10 epochs, and the weights after the last epoch are kept. Every variant uses this budget and checkpoint rule unless that is the change being tested.

B. Improvements (B)

Each variant changes one thing relative to its parent configuration. E1, class weights: cross-entropy weighted by $w_c \propto 1/n_c$, normalised to mean 1, so rare classes count as much in the loss as in balanced accuracy; re-weighting is analysed together with logit adjustment by Wang et al. [4]. E2, logit-adjusted loss: training uses $\mathrm{CE}(z + \tau \log \pi, y)$ with class priors π and $\tau = 1$, which is Fisher-consistent for the balanced error [5] and equals Balanced Softmax [6]; prediction uses the raw logits z. E3, flips and 90° rotations: dermatoscopy images have no canonical orientation. E4, TrivialAugment [7]: one random operation per image at random strength. E5, improved CNN: four blocks of two convolutions with BatchNorm [8], dropout 0.3 [9] before the head (1.18 M parameters). E6, cosine schedule [10] with two warm-up epochs. E7, pretrained ResNet-18 [11] with ImageNet [12] weights

(11.2 M parameters), fine-tuned end to end; inside the model the input is mapped to ImageNet normalisation and upsampled bilinearly to 112×112, so it still accepts the 28×28 input. MedMNIST+ benchmarks such models across resolutions [13]. E8, SAM [14] ($\rho = 0.1$) on top of E2, since SAM helps minority classes escape saddle points [15]; it is compared with E2. E9, post-hoc logit adjustment: the trained baseline's logits minus $\tau \log \pi$, with $\tau = 1.25$ chosen on validation data [5]. E10: 30 instead of 10 epochs, because the starter notes that the baseline is not finished after 10.

Stacking. Starting from the best single change (E2), we added the next change and kept it only if the mean validation balanced accuracy improved: S1 = E2 + ResNet-18, S2 = S1 + flips/rotations. E1 was not stacked with E2 because both correct the same imbalance. The final model averages the logits of the three S2 seed models in one TorchScript module.

C. Evaluation

Main metric: balanced accuracy; we also report macro-F1 and per-class sensitivity. Every variant was trained with seeds 0, 1 and 2 and is reported as mean ± std on the validation set. The 95 % interval of the difference to the baseline is a paired bootstrap [16]: 2 000 resamples of the 1 003 validation images, the same indices applied to all six runs, the difference of the seed-averaged balanced accuracies. The starter's bootstrap_difference compares one seed pair at a time; its intervals are wider and exclude zero for fewer seeds (one of three for E2; repository, results/starter_bootstrap.md), which is why we pool all seeds. All choices were made on validation data; the test split was evaluated once, after the final model was frozen. Runs were tracked per epoch and are viewed in TensorBoard (Fig. 3). Training ran on a laptop GPU (AMD, PyTorch MPS): about 1 minute per small-CNN run and 16–18 minutes per ResNet-18 run.

IV. Results

Table II and Fig. 1 compare all variants with the baseline. The three imbalance corrections (E1, E2, E9) improve balanced accuracy with intervals above zero; E3, E5, E7 and E10 show no clear effect alone; TrivialAugment and the cosine schedule are clearly worse. The stack S1 adds +0.043 over E2 and S2 a further +0.016 over S1; both intervals against the parent contain zero, while S2's interval against the baseline, [+0.078, +0.181], is the highest. SAM did not improve on E2 (−0.015 [−0.044, +0.014]).

Table III gives the single test evaluation. The final model improves balanced accuracy by 0.184 [+0.136, +0.232]. Dermatofibroma sensitivity rises from 0.09 to 0.57, melanoma from 0.34 to 0.64 and vascular lesions from 0.45 to 0.93, while nevus sensitivity falls from 0.91 to 0.73; macro-F1 changes little (0.479 to 0.500). Fig. 2 shows the same pattern on the validation set.

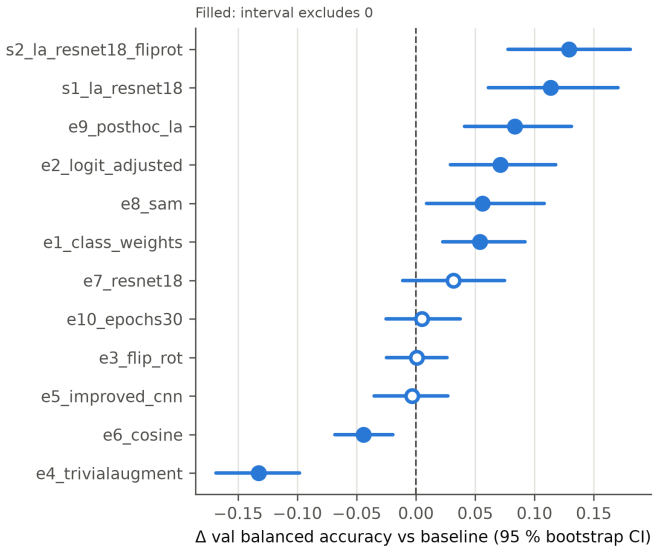


Fig. 1. Difference in validation balanced accuracy to the baseline with 95 % bootstrap interval (filled: interval excludes 0).

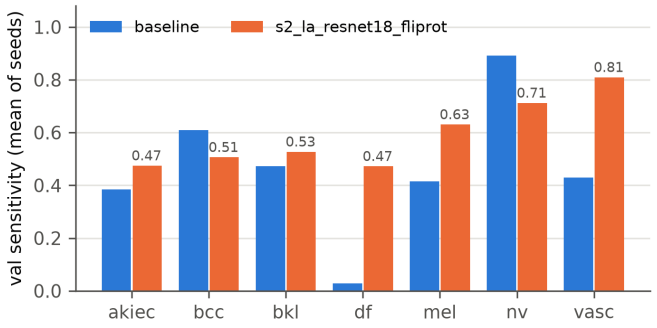


Fig. 2. Validation sensitivity per class (mean over seeds): baseline vs. S2.

V. Discussion

What worked. Every change that corrects the imbalance helped, as expected for a metric that weights the 80 dermatofibroma images like the 4 693 nevi. Logit adjustment did slightly better than class weights, consistent with its consistency for the balanced error [5]. The pretrained ResNet alone was not a clear gain, but with the logit-adjusted loss it gave the largest step: it kept nevus sensitivity at 0.76 (E2 alone: 0.58), so the imbalance correction over-corrected less and macro-F1 recovered (0.509 vs. 0.400). Averaging the three seed models added about two points.

What did not. TrivialAugment and the cosine schedule were clearly worse within the fixed 10-epoch budget. A likely reason for TrivialAugment is that strong augmentation needs more epochs; we did not test longer budgets for E4 or E6, so both explanations are hypotheses. Training three times longer (E10) did not help the small CNN, so training time is not what limits it; a larger plain CNN

TABLE II
Validation results, mean ± std over seeds 0–2. Δ: difference of seed-averaged balanced accuracy with 95 % paired bootstrap interval

	change	val bal. acc.	Δ vs. baseline [95 % CI]	Δ vs. parent [95 % CI]	val macro-F1
A	baseline (starter CNN)	0.461 ± 0.011	–	–	0.470 ± 0.007
E1	class-weighted cross-entropy	0.515 ± 0.016	+0.054 [+0.023, +0.092]	–	0.400 ± 0.022
E2	logit-adjusted loss, $\tau = 1$	0.532 ± 0.031	+0.071 [+0.029, +0.118]	–	0.400 ± 0.028
E3	flips + 90° rotations	0.462 ± 0.016	+0.001 [−0.025, +0.026]	–	0.480 ± 0.015
E4	TrivialAugment	0.329 ± 0.024	−0.133 [−0.169, −0.098]	–	0.367 ± 0.023
E5	BatchNorm, dropout, wider CNN	0.458 ± 0.047	−0.003 [−0.035, +0.027]	–	0.466 ± 0.050
E6	cosine schedule + warm-up	0.417 ± 0.016	−0.044 [−0.068, −0.019]	–	0.445 ± 0.018
E7	pretrained ResNet-18, 112 px	0.493 ± 0.051	+0.032 [−0.011, +0.075]	–	0.494 ± 0.052
E8	SAM on E2	0.517 ± 0.028	+0.056 [+0.009, +0.108]	−0.015 [−0.044, +0.014] vs. E2	0.375 ± 0.028
E9	post-hoc logit adj., $\tau = 1.25$*	0.545 ± 0.023	+0.083 [+0.041, +0.131]	–	0.364 ± 0.023
E10	30 epochs	0.466 ± 0.002	+0.005 [−0.025, +0.037]	–	0.490 ± 0.008
S1	E2 + ResNet-18	0.575 ± 0.035	+0.114 [+0.061, +0.170]	+0.043 [−0.018, +0.100] vs. E2	0.509 ± 0.044
S2	S1 + flips/rotations	0.590 ± 0.017	+0.129 [+0.078, +0.181]	+0.016 [−0.028, +0.054] vs. S1	0.478 ± 0.022
Final	S2, 3 seeds averaged	0.609 (ensemble)	–	–	0.505 (ensemble)

* τ chosen on the same validation images, so this row is slightly optimistic.

TABLE III
Test set, evaluated once after the freeze: both models are 3-seed ensembles. Per-class columns are sensitivities; Δ is a paired bootstrap on the 2 005 test images

model	bal. acc.	Δ [95 % CI]	macro-F1	akiec	bcc	bkl	df	mel	nv	vasc
baseline	0.448	–	0.479	0.33	0.55	0.46	0.09	0.34	0.91	0.45
final	0.632	+0.184 [+0.136, +0.232]	0.500	0.52	0.50	0.55	0.57	0.64	0.73	0.93

TODO: TensorBoard screenshot
(figures/tensorboard.png)

Fig. 3. Validation balanced accuracy per epoch for all runs in TensorBoard.

(E5) did not help either and was unstable across seeds (std 0.047). SAM added nothing on top of E2.

Trade-off and uncertainty. The gain in balanced accuracy is bought with lower nevus sensitivity (0.91 to 0.73 on test): more harmless moles are flagged. Benign keratosis, actinic keratoses and basal cell carcinoma stay near 0.5. With 12–33 validation images in three classes, intervals are wide, and with this many comparisons a few intervals may exclude zero by chance; the stacking steps over E2 are not individually significant. The test score (0.632) is above the validation score (0.609), within this noise.

Limitations and next steps. 28×28 inputs, one dataset, possible same-lesion images across splits, and a fixed 10-epoch budget for every variant. Next we would tune τ and the epoch budget for the ResNet, try deferred reweighting, and use the higher-resolution MedMNIST+ images [13].

VI. Conclusion

A logit-adjusted loss is the most effective single change for balanced accuracy on DermaMNIST, and it works best together with a pretrained ResNet-18 on upsampled inputs. Our final three-seed ensemble raises test balanced accuracy from 0.448 to 0.632, mainly by finding dermatofibroma, melanoma and vascular lesions, at the cost of nevus sensitivity.

AI use statement

We used Claude (Anthropic) through Claude Code throughout the project: it wrote most of the training, evaluation, export and app code, ran the experiments on our laptop, made the figures, and drafted the README and this report. [TODO: what the team reviewed in the code.] [TODO: confirm that the numbers were compared with results/summary.csv and results/test/.] [TODO: confirm that every reference was checked against the original paper.]

Contribution statement

TODO.

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